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Generating Music using AI

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Generating Music using AI

Generera Musik med AI

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Master's thesis work carried out at Axis Communications.

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Abstract

Network speakers are used in public spaces for public announcements and sometimes to play background music. To play music, a special commercial music licence is required, specifically made for use in public environments. However, customers have to pay licence fees and educate themselves on the topic of copyright licensing in order to do this according to current regulations. In this thesis we explored the possibilities of generating alternative, licence-free background music using machine learning methods. We surveyed the field for existing models and data sets, and carried out interviews with musicians to identify music quality characteristics that could be used as evaluation metrics.

We chose to tune and compare the transformer model GPT-3 and the Long Short-Term Memory (LSTM) model Performance_RNN. The music was evaluated using the COSIATEC algorithm to find recurrent patterns as well as using a custom metric based on Tymoszcko's theories of tonality. Experiments were carried out investigating the impact of learning rate, training data characteristics and generation parameters. Both GPT-3 and Performance RNN performed well at generating long term structure in music, but the training time and accuracy differed depending on the chosen data set. To add to the findings in this thesis it would be interesting to investigate the correlation between human perception of the music and the scores obtained in this report. It would also be of interest to further investigate the impact of the training data characteristics, such as genre and melodic content.

Keywords: MSc, Generative Model, Music generation, Transformer, Encoder, Decoder, VAE, Evaluation metric, Music quality

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Contents

1	Introduction	7
1.1	Project context	7
1.2	Aim	7
1.3	Methodology	8
1.4	Scope and limitations	8
1.5	Previous work	8
1.6	Report outline	9
2	Background	11
2.1	Machine learning basics	11
2.1.1	Artificial Neural networks - ANN	12
2.1.2	Optimization aspects	12
2.1.3	Transfer learning	14
2.2	Generative models	15
2.2.1	Autoencoder - AE	15
2.2.2	Transformers	15
2.2.3	Long Short-Term Memory - LSTM	17
2.2.4	Examples of generative models	17
2.3	Music Theory	19
2.3.1	Fundamentals	19
2.4	Digital music representation	21
2.4.1	MIDI (Musical Instrument Digital Interface)	22
2.4.2	Raw audio	22
2.4.3	Text representation	23
3	Method	25
3.1	Literature review	25
3.2	Interviews	26
3.3	Metrics	26
3.4	Music generation	28

3.5	Training data	28
3.5.1	ABC format parsing in GPT-3	29
4	Results	31
4.1	Interview results	31
4.2	Dataset evaluation results	32
4.3	Pretrained model results	33
4.4	Magenta: Performance RNN	34
4.4.1	Learning rate	34
4.4.2	Seeding sequence	34
4.5	GPT-3	36
4.5.1	Prompt	36
4.5.2	Learning rate	37
4.5.3	Epochs	38
4.5.4	Transposed training data	39
4.5.5	Temperature	40
5	Discussion	43
5.1	Model performances	43
5.1.1	Data set homogeneity	43
5.1.2	Learning rate	44
5.1.3	Epochs	44
5.1.4	Temperature	44
5.1.5	Data format parsing	44
5.2	Metric	45
5.2.1	Note density	45
5.2.2	Empty beat ratio	45
5.2.3	Consonance	45
5.2.4	Centricity	46
5.2.5	TECs	46
5.2.6	Macroharmony	46
5.2.7	Groove	47
5.3	Legal aspects	47
6	Conclusion	49
	References	51
A		55
A.1	Interview documents	57
A.1.1	Consent to be interviewed for research	57
A.1.2	Interview Protocol	58

Chapter 1

Introduction

Artificial Intelligence (AI) generating music has had a significant acceleration of progress during the last decades [11] [31]. In generative modelling an AI model is trained to generate new data such as images, text or audio. This thesis deals with the generation of score-based music.

1.1 Project context

This master thesis project is carried out at Axis Communications, a company providing network connected security solutions. They are most known for their security cameras, but Axis also provides network connected speakers for commercial use. The speakers are used in public spaces such as malls, train stations, airports, etc, to give public announcements or in some cases to play background music. In this thesis AI generated music is researched because it can be licence free. This is further discussed in Section 1.4. Music generated using AI can therefore act as a an alternative to a commercial licence when playing music in a public space. Axis wants to explore the possibility of providing licence-free music that their customers could use freely.

1.2 Aim

The aim of this thesis is to explore ways of generating music using machine learning which is suitable for use as background music. There has been great success in the field of generating music using AI in recent years, with producing convincing piano melodies [23][31] and even music with realistic vocals [11]. In the context of this thesis, the music in question will be played in public environments. The focus will therefore be on generating music that is suitable as background music in public spaces. More specifically, this thesis will explore the possibilities in music generation by investigating existing methods and models, testing out

well-supported choices of models, finding objective ways of evaluating and comparing the methods, and ultimately to find a method that works for the specific context at Axis. The work in this thesis has been driven by the following research questions:

- RQ1 Which already developed models in the state-of-the-art can be utilised to achieve music generation?
- RQ2 What are the demands on the quality of background music in public spaces?
- RQ3 How can the quality of generated music be measured?
- RQ4 To what extent can the existing generative models of background music generate background music suitable for public spaces?

1.3 Methodology

The work in this thesis was carried out using a mixed methodology approach [25] consisting of reviewing the state-of-the-arts, conducting semi-structured interviews and carrying out experiments. The results of the experiments were evaluated from a quantitative perspective.

1.4 Scope and limitations

The end-results of this project are envisioned to be used in a commercial context. This requires a concern for legal issues in choosing a training data set and using pretrained models. Many AI models are publicly available under a creative commons licence but not always allowing commercial use of the software. Using copyrighted music for training could also be problematic for legal reasons. However, today there are plenty of music that has fallen into the public domain that can be used freely. Therefore copyright-free music will be used for training as much as possible.

The result of this thesis is also limited by the available computing power in combination with time constraint, as is with many machine learning project. The most successful generative models have been trained on millions of data samples over a long period of time on strong computers, wherefore this particular setup will not be able to reach the same results. Focus will instead be put on the comparison of different models, both pretrained and retrained.

Because of the breadth of the field of generative AI modelling, the models chosen for further investigation were limited to only transformer or Long Short-Term Memory (LSTM) models.

There are other potential applications for AI generated music that will not be covered in this thesis, for example as a tool for musicians in the composing process or an unbiased way of finishing unfinished work by late composers [12][8].

1.5 Previous work

This work is partially based on the work by Gonsalves [14] in which GPT-3 was retrained using the OpenEWLD data set in ABC style music notation.

1.6 Report outline

The content of the report is structured as follows:

Chapter 1 - Introduction: Introduces the subject, context and aim of the thesis. The scope and its limitations are introduced together with the research questions.

Chapter 2 - Background: Contains necessary theory explained concerning machine learning, music, digital processing of music and music evaluation metrics. The background is intended to give all the necessary information to be able to fully comprehend the method and results.

Chapter 3 - Method: This chapter includes a detailed description of how the work was conducted in researching the research questions.

Chapter 4 - Results: The results of the carried out experiments. This section also contains a brief evaluation of the results.

Chapter 5 - Discussion: A detailed analysis of the results and the experiment setup.

Chapter 6 - Conclusion: This section contains a conclusion answering the posed research questions.

Chapter 2

Background

This chapter describes the background theory for this thesis. In Section 2.1 fundamental machine learning theory is introduced, Section 2.2 handles some generative modelling architectures, Section 2.3 gives an introduction to music theory and finally some details about processing music digitally are presented in Section 2.4.

2.1 Machine learning basics

Machine learning is a subfield of Artificial Intelligence. Machine learning algorithms mimic the human learning process by observing large amounts of data and learning the patterns in them. In general terms, machine learning aims to implement algorithms that are able to learn from experience. This means that the system does not need to be explicitly programmed, which enables complex tasks to be solved with less complex code. There are many different types of machine learning models and many ways of categorizing them. In this project it is especially important to distinguish between discriminative and generative models.

Discriminative models A discriminative model is capable of classifying data. More specifically this means finding the probability that data with a certain set of characteristics belong to a specific category, a *class*. As an example, a good discriminative model could be able to classify the contents of an image or the genre of a song. Discriminative models are used in practice in face recognition, self-driving cars, medicine and many other fields.

Generative models A generative machine learning model can classify data, just like a discriminative model, but with the additional capability of being able to generate new data. If a discriminatory model predicts how likely data with a certain set of characteristics belong to a certain label, the generative model can predict the probability of a certain set of characteristics, given the label. This means that when a generative machine learning model is given the instruction to generate a picture of a dog, it generates it based on the characteristics

it learned through classifying images of animals. In practice it may for example find the probability of a dog having fur and four legs, and then generates the image based on these findings.

2.1.1 Artificial Neural networks - ANN

An artificial neural network is a machine learning algorithm that models the neurons and synapses of the brain. The neural network architecture can be seen in Figure 2.1. A neural network consists of multiple node layers, where each node receives input from the nodes in the previous layer and passes on its output to the next layer. Based on this, the network can be modelled as a directed graph which is why the neurons are sometimes referred to as nodes. In each node the input is evaluated by the nodes activation function to calculate the node output, as shown in an example in Figure 2.2. The standard activation function adds weights to the inputs, sums the weighted values and adds a bias. The weights of the activation function are updated each time the network has been propagated through, the network "learns" the weights. The training process can either be *supervised* or *unsupervised*.

In supervised learning, the training data is labelled. The label, i.e the desired output, is compared to the output of the network. Depending on how close the network is to making the correct prediction the weights are adjusted accordingly.

In unsupervised learning in neural networks, the training data is unlabelled. The network does not know anything about the data, and instead learns patterns in it. For example the network can group cats and dogs into two different groups based on their characteristics, but it does not know what it is categorizing. In unsupervised generative models it is possible to generate random pictures that could end up being a picture of a dog, but one could not ask the model to explicitly generate a picture of a dog since it does not know *what* it modelled.

Feed-forward Neural Network

The feed-forward neural network is the simplest type of ANN. It does not form any cycles, which means that the information is sent in only one direction in the graph.

Recurrent Neural Network - RNN

In a recurrent neural network the outputs from the nodes are sent back as input to the node or into previous node layers. An advantage of RNNs is that they can predict sequential data, which the standard feed-forward neural network cannot.

2.1.2 Optimization aspects

Learning rate

The learning rate controls how fast the neural network learns. It does this by determining how much the weights should be updated in each training iteration. If the learning rate is too small the training process gets very long, but if the learning rate is too big the model could converge too quickly and not give good enough results.

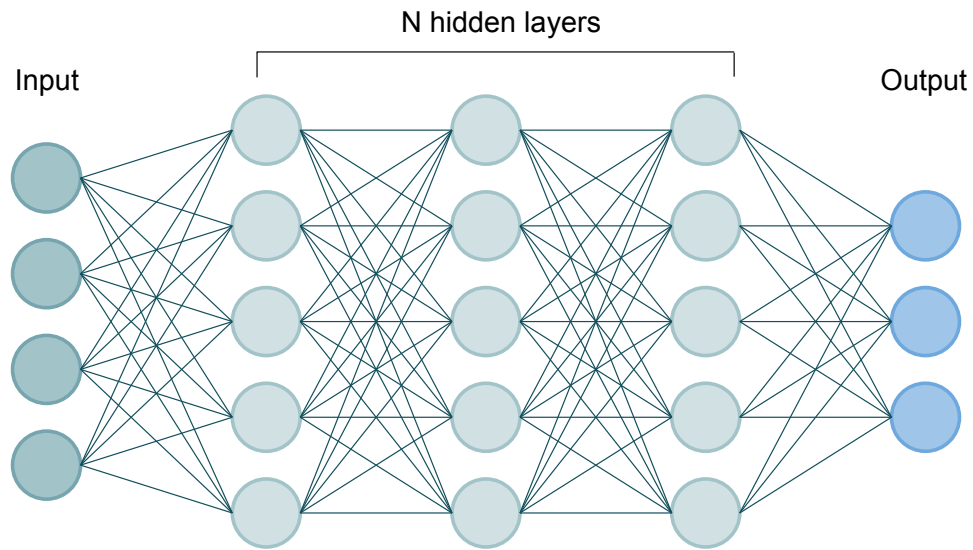


Figure 2.1: An example of the Artificial Neural Network (ANN) architecture. The ANN consists of an input layer, N hidden layers and an output layer of different sizes. The hidden layers can have different sizes which affects the performance of the model. The input layer size corresponds to the size of the input data sample. The output layer size corresponds to the size of the desired output sample size or the number of possible classes in the case of a classification application.

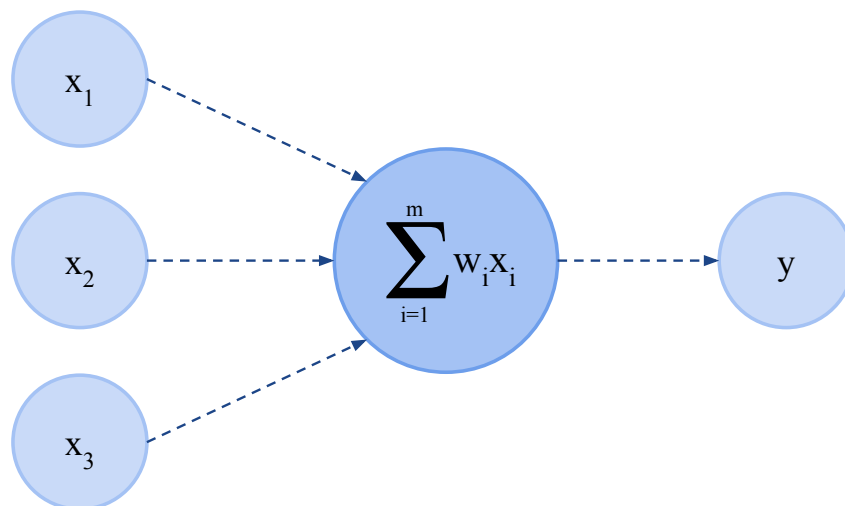


Figure 2.2: The standard activation function and how it evaluates the input to determine the output. A bias term can also be added to the output.

Loss

An ANN learns by minimizing the *loss function*. The loss function is the difference between the true desired output and the network output. A common loss function to use for sequential data is the cross entropy loss function [13]. Cross entropy is defined in Formula 2.1. $p(x_i)$ denotes the original probability distribution and $q(x)$ is the predicted distribution, the output of the model.

$$H(p, q) = - \sum_{x_i} p(x_i) \log q(x_i) \quad (2.1)$$

Batch size

The batch size defines the number of data samples used in one training iteration before updating the weights. A large batch size in the size of the data set can lead to the model being bad at generalizing data it has not seen before. The optimal batch size to use is different for every model and data set.

Temperature

The temperature is a parameter used when generating data samples. It determines the confidence of the model, where a low temperature benefits the most probable results and therefore generates data with less diversity. Instead using a high temperature includes less probable sample generations, resulting in more variation in the data but also more mistakes.

2.1.3 Transfer learning

To fine-tune an already trained model to fit your specific problem is called *Transfer learning*. Training a new model from scratch is time- and resource consuming and generally only available to big companies and organizations with enough funds [24]. In the case of music generation, there are many models that have been trained on classical music. Even though the model is genre-specific, it will have learned basic, low-level musical patterns that can be useful when generating music in other genres. Transfer learning could for example mean that the majority of the learned weights of the trained models are kept, while some of the last layers are retrained on new data. This enables high accuracy without having to train the entire model.

There are extensively trained models publicly available whose results are almost impossible to replicate with one GPU even if large time resources were available. The transparency and mutability of the models vary, where the models are sometimes available without insight into the source code and therefore have limited room for customizing. In some models all of the trained weights are available, and can therefore be fine-tuned through transfer learning to tailor the model to a specific problem.

2.2 Generative models

This section describes some common architectures used for data generation. Focus is put on models that perform well with sequential data such as audio.

2.2.1 Autoencoder - AE

An autoencoder is a type of neural network that learns how to represent data in a meaningful way using less information than the original representation. The autoencoder architecture consists of an encoder network and a decoder network. The data is encoded and then represented in the latent space, z , which is why these types of architectures are sometimes called deep latent variable models. After the data have been encoded, the aim is to reproduce the original data by decoding the encoded representation. Figure 2.3 shows the autoencoder architecture in more detail. The figure visualizes how the input is encoded in the latent space by using an example from the MNIST data set [10]. In the latent space the image is clearly represented using less information, in this case fewer image pixels. It can then be decoded to reproduce the original input. Because of the shape of the network architecture, the latent space is often called a bottleneck.

The encoder and decoder of the autoencoder architecture can be used together or separately as different parts of a generative model. By encoding data using an encoder, one can benefit from the data being represented with less information by using this as training data. This can make training processes faster since a big part of what makes training take time is the information heavy data. After training, the output can be decoded by a decoder in order to get a sample with the sought after detail [4].

Variational Autoencoder - VAE The variational autoencoder (VAE) is similar to the autoencoder, but represents the input data as a normal distribution in the latent space. The practical difference from the regular autoencoder is an additional layer where the compressed data in the latent space are normalized. The distribution is called the *true distribution* of the data.

The additional normalizing layer means that the representation in the latent space is continuous, which makes sampling from the latent space more accurate than with the regular autoencoder. A drawback of VAEs is that they tend to have difficulties with sequential data and modelling long-term structure [23]. Furthermore, the information loss that in many ways is the big advantage of the model is also a compromise for quality as the generated data is often noisier than the original data [4].

2.2.2 Transformers

The transformer architecture was introduced by Vaswani et. al in 2017 [29]. The transformer architecture has an encoder-decoder structure and an inherent attention mechanism that enables it to perform very well with sequential data with long-term structure. Because of this it does not rely on recurrence which makes it faster to train than other encoder-decoder models.

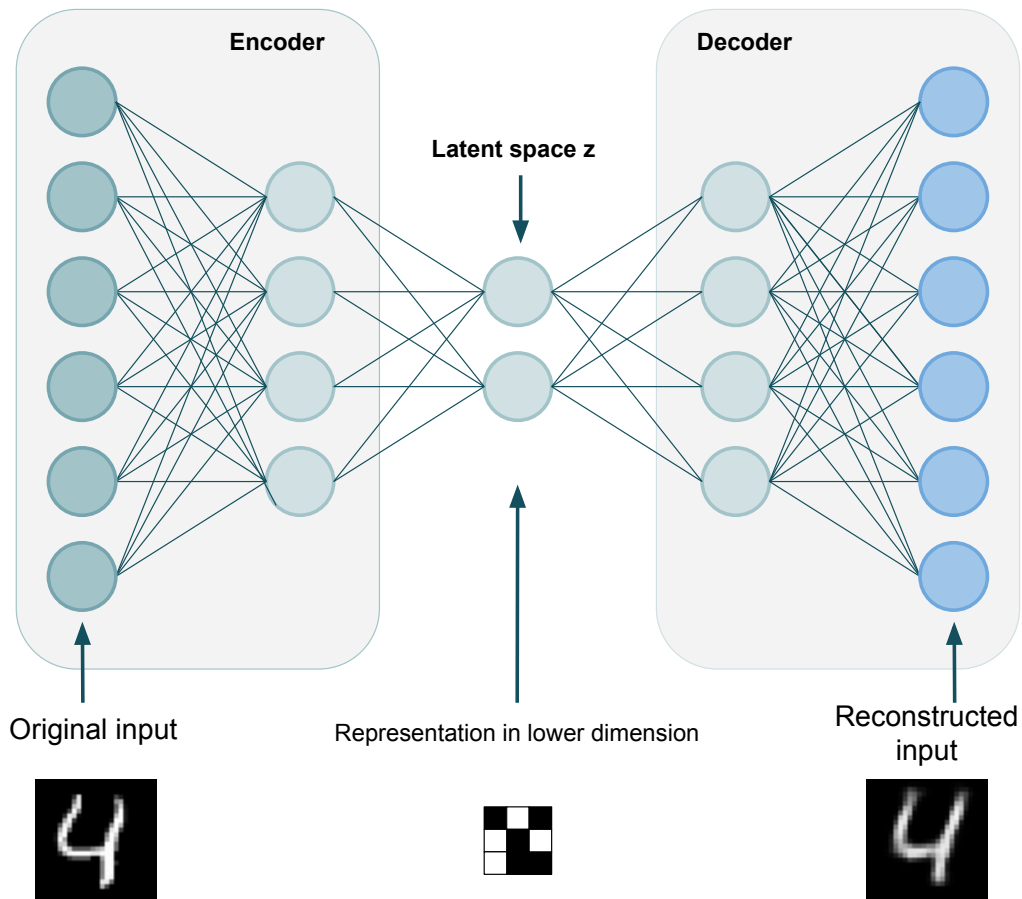


Figure 2.3: The autoencoder architecture consisting of an encoder, a decoder and the latent space representation, or bottleneck. In this example the autoencoder encodes an image from the MNIST dataset [10] and decodes it into a noisier reproduction.

The attention function is presented in Equation 2.2 and takes queries ($\mathbf{Q}$), keys ($\mathbf{K}$) and values ($\mathbf{V}$) as input and maps them to an output value. d_k is the dimension of the $\mathbf{K}$ and $\mathbf{Q}$ vectors and this factor scales the dot-product to avoid vanishing gradient problems, which is a common issue with neural networks where the gradient of the loss function becomes too small resulting in an undertrained network. $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ are all derived from the same input when referring to self-attention.

$$\text{attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (2.2)$$

Attention Attention is the ability to identify context. The correct understanding of a word is important in a sequence completion task, both for language and music. The following two sentences illustrate this in a language example by showing two very different meanings of the word *cross*:

The animal didn't cross the street because it was too tired.

You don't want to cross them.

The meaning of a word in a sentence is deduced by its surrounding context, and can even depend on tokens from further away in the sequence than in the direct proximity of the word. Figure 2.4 shows how the word *it* relates to the other words in the sentence. It depends the most on the words *the animal*. If there were instead multiple animals not crossing the street, it would have been replaced by *they*. The word *it* depends directly on some words, and contextually on others. The same theory applies to music, which also has structural semantics similar to language.

The transformer architecture can use the whole previously generated sequence to predict the next. In comparison, an RNN forgets tokens from further back in the sequence and only takes recent words, or as in the context of this thesis, music notes, into account.

2.2.3 Long Short-Term Memory - LSTM

The LSTM model architecture is a type of Recurrent Neural Network that was developed in response to the common vanishing gradient problem with regular RNNs [17]. An LSTM has a feedback loop from previous layers in the network which is why it is said to "remember" previous values. Compared to the regular RNN network the LSTM remembers values further back in the network, which is why it performs better with sequential data, and also avoids the vanishing gradient problem.

2.2.4 Examples of generative models

The following models are examples of the architectures described in Section 2.2.1, Section 2.2.2 and Section 2.2.3 and are used to investigate music generation in this thesis.

MusicVAE is a variational autoencoder model developed by the Magenta Tensorflow team in 2018 [23]. It uses a two-dimensional LSTM encoder and a hierarchical LSTM decoder.

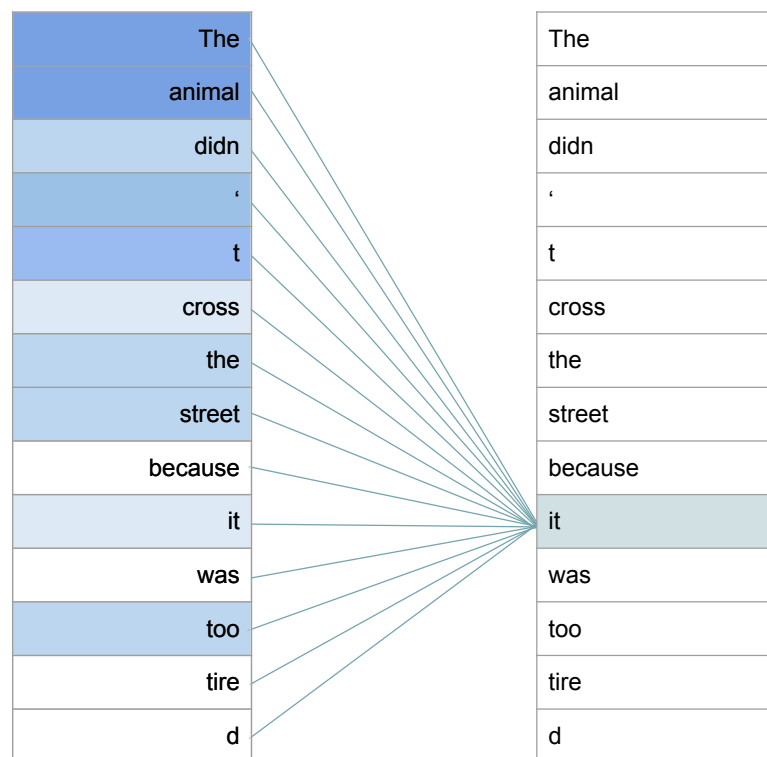


Figure 2.4: Illustration of the attention mechanism. Attention deduces how the words in a sentence inherently relate to each other on the basis of context.

LSTM networks are introduced in Section 2.2.3. Input data of dimension 2048 was encoded into 512 latent dimensions. This model has freely available trained weights and was trained on a MIDI dataset created by the Magenta team themselves by collecting 1.5 million songs from the internet. Instead of directly decoding from the latent space, decoding is done bar-wise by using a conductor for each bar and decoding from these separately, with the aim of improving performance with sequential data.

GPT-3. GPT stands for Generative Pre-trained Transformer and GPT-3 is the third generation GPT model. GPT-3 was at the time of its release 10 times bigger than any other sparse NLP model with its 175 billion parameters [7]. GPT-3 was trained on 45 TB of text data collected from the internet, and is designed for use in NLP tasks, in which it achieves high accuracy. This model can be applied to music generation by representing music in a text format. By fine-tuning the pre-trained GPT-3 model it can be used to generate music while still taking advantage of the sense of context GPT-3 already has.

MuseMorphose is a transformer-based variational autoencoder model developed by Taiwan AI Labs in 2021 [31]. The developers had the aim to create a model that allowed the user to have an impact on the rhythmic intensity and polyphony of music. The model both generates music from scratch, but can also take a song as an input and recreate it based on the conditioning parameters. MuseMorphose was trained on 10 626 songs from the Remi pop data set.

Performance RNN is an open source LSTM model for music generation from Magenta Tensorflow [26]. The model focuses on generating music with expressive timing and dynamics, and successfully generates cohesive music with local long-term structure. It is trained on the Yamaha e-Piano competition dataset containing 1400 midi files.

2.3 Music Theory

Music theory [6] defines a set of principles that music we like to listen to tends to follow¹. A big part of music theory is rooted in the physics of sound as well as in complex mathematics, but an equally big part is the more intangible that comes from human intuition and creativity. A general convention is that popular music follows music theory rules, but in practice a lot of great music also breaks them. Deviating from the convention requires knowledge of how much and in what way to do this. This is a much more difficult modelling problem than just sticking to the clear rules of music theory, which is why the objective in this thesis is that the generated music should follow conventional music theory.

2.3.1 Fundamentals

In this section some fundamental music theory is presented. It is later used in the evaluation of music and is an important aspect in finding a quantitative metric. The theory is explained based on the piano clavier, seen in Figure 2.5.

¹Note that when referring to music theory in this report, we generally mean western music theory.

Pitch The frequency of a sound wave corresponds to its pitch. The higher the pitch, the higher the frequency.

Tones Western music is based on 12 different tones: C, C#/Db, D, D#/Eb, E, F, F#/Gb, G, G#/Ab, A, A#/Bb and B. The notes with # or b in front of them are called sharps and flats respectively and correspond to the black keys on a piano. The notes without # or b correspond to the white keys. The distance between each of the twelve notes is a determined distance called a semitone.

Octave The next double frequency of a tone marks an octave. For example, 220 Hz, 440 Hz and 880 Hz all correspond to the note A but in different octaves. We say that they are harmonics of the same tone. This means that the same set of tones are repeated in octaves spanning over the entire piano in Figure 2.5.

Harmony Harmony is one or more notes played at the same time. Three or more notes form a chord. Playing two or more tones simultaneously can either sound dissonant or consonant depending on the note distance between the notes. Dissonant sounds feel unresolved and can be stressful or uncomfortable to listen to. Consonant sounds are the opposite and sound pleasant. Consonance and dissonance can be used in alternation to create tension and emotion in music.

Keys Western music usually follows a key. A key defines a set of eight tones which are distanced with a decided distance from each other so that no dissonances appear. The *major* and *minor* keys are most commonly used in western music but there are also other modes. The eight white keys make up the C major key, but the major key can be transposed to have any note as the base. Transposing a key does not change the melodic content, only the pitches.

Every major scale has a parallel key in minor. This key contains the exact same tones but has another tone as the base. The base tones of the two parallel keys are spaced three semitones from each other, e.g the parallel key of C major is A minor.

Some notes in the key have a greater attraction to the base tone than others. These are the fourth and fifth tones. In many popular songs the chords corresponding to these specific notes are used more than the others in the key.

Repetition An important part of music is the rhythm and consistent beat as well as repetition to some extent. Music with too much and too little repetition can both be uncomfortable to listen to.

Tonality Tonality is the principle of arranging music around a central note, a tone. More specifically it defines a set of rules for the specific relationship between chords, notes and keys that regulate much of modern and older music [1]. We can call music *tonal* or *atonal*. In the following sections the characteristics of tonal music are presented.

Criteria for tonal music In this project tonality is of interest because essentially all music we listen to is considered to be tonal, and it is therefore desirable that the music we generate also is tonal. Tonality is also a widely discussed concept that has been theoretized

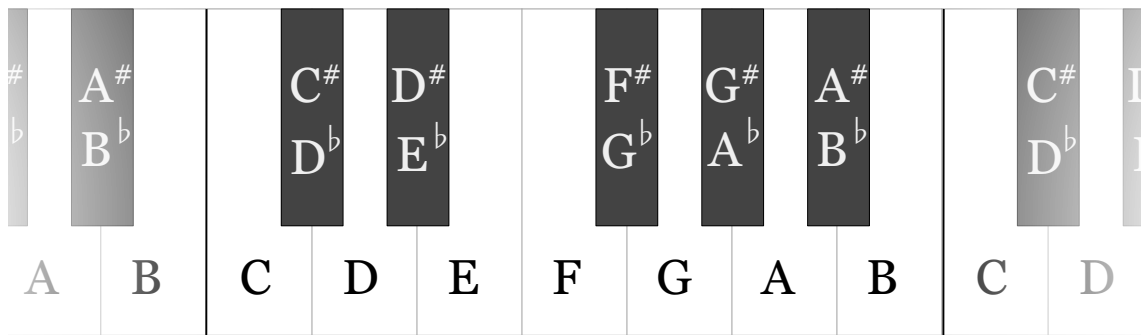


Figure 2.5: An octave on a piano. The octave is repeated multiple times and the 12 tones in the octave can be played in different pitches.

in such a way that it could act as support for a potential evaluation metric.² Dmitri Tymoczko proposes a set of criteria [28] that define the characteristics of tonal music. The five proposed criteria that a tonal musical piece contains are as follows:

- **Conjunct melodic motion.** Melodies tend to move by short distances from note to note. To illustrate, the distance between C and D is shorter than the distance between C and G in terms of note distances.
- **Acoustic consonance.** Consonant harmonies are preferred to dissonant harmonies and tend to be used at points of musical stability.
- **Harmonic consistency.** The harmonies in a passage of music tend to be structurally similar to one another over time.
- **Limited macroharmony.** Tymoczko uses the term “macroharmony” to refer to the total collection of notes heard over moderate spans of musical time. Tonal music tends to use relatively small macroharmonies, often involving five to eight notes.
- **Centricity.** Over moderate spans of musical time, one note is heard as being more prominent than the others, appearing more frequently and serving as a goal of musical motion.

2.4 Digital music representation

Music can be represented digitally in different formats. The different file formats can contain large amounts of information, or be more stripped-down and potentially lack detailed information. This impacts the final sound and modifiability of the sound file.

²Here’s an attempt: <https://github.com/sebasgverde/music-geometry-eval>

2.4.1 MIDI (Musical Instrument Digital Interface)

MIDI is a common file format to use because of its portability and widespread use. It is a score-based representation, meaning that a MIDI file contains instructions for how the music should sound. It then has to be parsed and converted into a raw audio file format to be listenable. An example of a MIDI file visualization can be seen in Figure 2.6. MIDI contains information about the velocity, pitch, vibrato, instrument and length of each note. It also contains information about the entire track such as key, tempo and instrument.

The MIDI file format is generally of small size and is flexible since it is easy to change instruments and the general composition of the piece. In the machine learning context this means that music generated in a MIDI format can be post-processed using a music processing software. A drawback of MIDI is that each instrument has to be generated separately. Another drawback is that the MIDI format has a determined set of possible note onsets, which is why there is no room for onsets outside of these. This can be compared to a continuous set of note onsets. The limited onset resolution can result in information being lost when converting between file formats because notes are placed in the closest onset space. The onset resolution is determined by the number of pulses per quarter note (PPQN) which is included in every MIDI file.

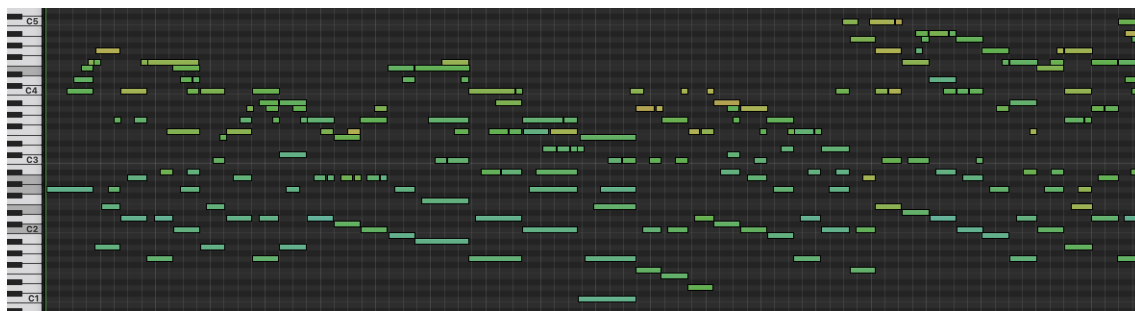


Figure 2.6: An example of a MIDI file. Pitch is shown on the x-axis and time is shown on the y axis. The velocity of each tone is represented by the colour of each bar in this particular visualization. The note length corresponds to the bar lengths of each note.

2.4.2 Raw audio

Raw audio formats such as wav and aiff contain the actual sound waves of the music. This means that they are many times more information heavy than MIDI. This is a limitation in that it demands a lot more computational power, but it also has more possibilities because of the added information. In comparison with language, MIDI would be a text string while raw audio is a sound file of the text being read out loud. With raw audio even singing can be learnt, and the model can learn to replicate the personal singing voice of a famous artist by completely copying the frequency spectra of their voice [11].

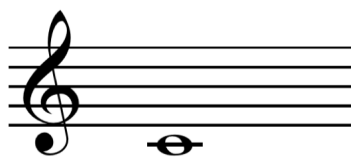
2.4.3 Text representation

Music can also be represented using text. This means that natural language processing models can be used for training, since these were trained on text.

MusicXML is an XML-based text representation of music. It contains information about the key and song title and presents each note, its duration, pitch and position in the song. This notation does not contain a lot of information compared to for example raw audio but has a lot of extra XML syntax elements why it gets very long even for short music pieces.

ABC notation ABC notation contains a header and a part containing the musical contents of the song. The header contains information about the key, time, default note length and song name. ABC notation uses letter notation, a-g and z to denote the note values and rests. Other elements are used to mark note lengths, chords and sharps and flats. The notes are described in their designated order, why the position of each note does not have to be explicitly written out.

Figure 2.7 shows the difference in length between MusicXML and ABC notation for the same piece. MusicXML is much wordier without giving any extra information about the piece. See Section 3.5.1 for more details about the formats used in this report.

**ABC notation:**

X: 0
T: Middle C
M: 4/4
L: 1/4
K: C
C4 |

MusicXML:

```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<!DOCTYPE score-partwise PUBLIC
  "-//Recordare//DTD MusicXML 3.1 Partwise//EN"
  "http://www.musicxml.org/dtds/partwise.dtd">
<score-partwise version="3.1">
  <part-list>
    <score-part id="P1">
      <part-name>Music</part-name>
    </score-part>
  </part-list>
  <part id="P1">
    <measure number="1">
      <attributes>
        <divisions>1</divisions>
        <key>
          <fifths>0</fifths>
        </key>
        <time>
          <beats>4</beats>
          <beat-type>4</beat-type>
        </time>
        <clef>
          <sign>G</sign>
          <line>2</line>
        </clef>
      </attributes>
      <note>
        <pitch>
          <step>C</step>
          <octave>4</octave>
        </pitch>
        <duration>4</duration>
        <type>whole</type>
      </note>
    </measure>
  </part>
</score-partwise>
```

Figure 2.7: Comparison between MusicXML and ABC notation of a simple song only containing a whole middle C note.

Chapter 3

Method

The purpose of this chapter is to describe the detailed approach used to address the following research questions (also listed in Section 1.2):

- RQ1** Which already developed models in the state-of-the-art can be utilised to achieve music generation?
- RQ2** What are the demands on the quality of background music in public spaces?
- RQ3** How can the quality of generated music be measured?
- RQ4** To what extent can the existing generative models of background music generate background music suitable for public spaces?

To address **RQ1** a review of the state-of-the-art was done through a semi-structured literature review. Investigating music quality in **RQ3** was done through interviews with professional musicians. The results from the interviews in combination with a semi-structured literature review addressed **RQ2**. Finally, experiments were carried out to address **RQ4** where models found in **RQ1** were implemented using metrics from **RQ3**.

3.1 Literature review

To find relevant studies in order to answer **RQ1**, a semi-structured literature search was conducted [19]. We chose the database Google scholar to conduct the search. Google scholar contains papers from many different databases and publishers which is why it is a good choice in order to avoid bias in favour of any specific publisher. To find relevant papers, a set of search strings were specified.

The articles were chosen based on a set of criteria that specify the relevance for the specific topic.

Search string RQ1: music generation OR generative model OR transformer music

Table 3.1: Inclusion and exclusion criteria for RQ1.

Inclusion criteria	Articles chosen
1. The paper presents a generative model architecture that is foundational research in the field	4
2. The paper presents a novel generative model architecture	10
3. Secondary and tertiary studies that review or map the current state-of-the-art in generative AI modelling	1
Exclusion criteria	Articles excluded
1. The model implementation is not publicly available	
2. The generative model type is not transformer- or LSTM-based	2
3. The paper deals with generation of raw audio	2
Papers found:	11

In addition to the semi-structured literature review, articles were also found through citation analysis, or snowballing [19], which is a reason for its semi-structured nature. This was done by back-propagating the sources in some of the articles found through the literature review. Snowballing was used when searching for papers about music evaluation criteria for **RQ3** by investigating what metrics had been used in evaluating the models in the articles from **RQ1**.

3.2 Interviews

Four semi-structured interviews [25] were carried out with professional musicians. They had all composed music of their own and had rigorous musical education. The aim of the interviews was to attempt to define the more abstract aspects of what makes music sound good, and later use this knowledge either to create or choose a mathematical evaluation method. The analysis was carried out by finding patterns and common themes in the interviews. The interview protocol and consent form is included in Appendix A.1.

3.3 Metrics

The interviews together with the literature review resulted in a set of important, measurable characteristics of music. Most of the metrics do not have a specified reference value, but instead need to be compared to a reference piece or distribution. To find reference points for the generated music, metrics were computed for the training data sets, which were used as a representation of human-made music. This shows how the scores vary and can give a hint of what could be a lower limit score or an interval to stay within. The properties of the music were evaluated using the following metrics:

Consonance. This metric is calculated as the key consistency of a song, which is measured by registering to what extent the notes that are played in a sequence belong to the same key. The consonance is measured between 0-1 for each key in major and minor, meaning that a song in C major should get a consonance score of 1 for C major and 0 for every other key. A high consonance score is therefore considered as an indicator of good quality.

Centricity The harmonies in a musical piece should be structurally similar, meaning that some notes and chords are more common than others. This can be measured by collecting the occurrence of each note in a histogram, whereafter the statistical entropy of the histogram is measured. Entropy is high if all notes are common, and low if a couple of notes dominate. It is therefore assumed that a song with low entropy follows tonality theory better [30].

Macroharmony The total number of used notes in the song is defined as the macroharmony. The maximum macroharmony is 12, i.e. notes in different octaves count as the same note. Tymozskos theory of tonal music (presented in Section 2.3.1) say that tonal music should have a macroharmony of 5-8 notes [28].

Groove Grooving pattern similarity, or consistency of rhythm. We limit the generated music by saying that the rhythm in bars close to each other in time should not differ too much but be somewhat consistent. For music in general this is not always true, but to branch out when it comes to rhythm requires more detailed specification of how this should be done [30].

Empty beat ratio The music should not have completely silent parts in the middle of the sequence, which is why a low empty beat ratio is aimed for.

Note density The note density measures the ratio of note onsets, i.e. the number of note onsets divided by the total length of the sequence. Previous research [5] [16] shows that a low song tempo can have a calming effect on the listener. Lowering the tempo makes the note density lower for a sequence length fixed in time, which then results in note density correlating with tempo. With this reasoning the aimed at note density should not be too high.

Compression ratio and TECs The structure induction algorithm (SIA) by Meredith [21] can be used to calculate repetition and patterns in long-term structure. SIA identifies translational equivalence classes (TECs) by representing each note as a point and applying a pattern recognizing algorithm to it. From this a visual representation of the musical patterns can be obtained. SIA finds all patterns in a musical passage, including ones that are not of interest from a musicological perspective. COSIATEC by Meredith [21] is an improved implementation of SIA and finds actual musical patterns more accurately by filtering out the most relevant patterns based on coverage, compactness and compression ratio. The compression ratio measures the total number of points that make up a pattern's total number of occurrences.

Table 3.2: Details about type, training data and licence for the models chosen for evaluation.

Model	Type	Pretrained on	Licence
GPT-3	Transformer	45 TB of text data (Common Crawl dataset)	OpenAI licence (Off-the-shelf licence, if purchased commercial use is permitted)
Performance RNN	LSTM-based RNN	Yamaha e-Piano Competition dataset (~1400 songs)	Apache 2.0 (Commercial use permitted)
MusicVAE	VAE	1.5 million midi files from the web [23]	Apache 2.0 (Commercial use permitted)
MuseMorphose	Transformer-VAE	Remi Pop-dataset (10 626 songs)	MIT (Commercial use permitted)

3.4 Music generation

Four models were chosen for evaluation, presented in Table 3.2. The models chosen for evaluation are GPT3, Magenta Music_vae, Magenta Performance_rnn, and MuseMorphose. Pre-trained weights for music generation were available for all models except GPT-3 since it is an NLP model. The models were first run using these pretrained weights, and then retrained using new datasets. In all iterations the training dataset as well as the generated music was evaluated for comparison. When evaluating the training data the full dataset was evaluated, whereas for the generated music 100 samples from each model iteration were used for evaluation.

The models were chosen based on model type, documented performance, as well as customizeability. All the models perform well on sequential data with long-term structure, which gives better results when generating music. No raw audio models were tested because of the extensive training time and resources needed. All the chosen models were either trained on MIDI or text representations, which facilitated training the models on the same data set.

After evaluating the generated music based on the pretrained weights, GPT-3 and Performance RNN were chosen for retraining and optimization. This was based on the results presented in Chapter 4 as well as the fact that we wanted to explore the possibilities of music generation with GPT-3. The two models were also of two different types, LSTM and Transformer.

3.5 Training data

Four data sets were found that could potentially be used for training. These were all evaluated using the same metrics used for the generated music. The four potential data sets are presented in Table 3.3.

Table 3.3: The datasets used for training.

Dataset	Size	File format	Public domain	Comment
EMOPIA [18]	1087	MIDI and REMI	No	Piano music collected from youtube. Licenced under creative commons attribution licence.
MAESTRO	1276	MIDI and wav	Yes	Classical music in the public domain
OpenEWLD [27]	502	MusicXML	Yes	Derived from EWLD to only contain music in the public domain
Trumpet Kings of the Swing Era	177	Multitrack Midi	Yes	Small dataset but contains multiple instruments

3.5.1 ABC format parsing in GPT-3

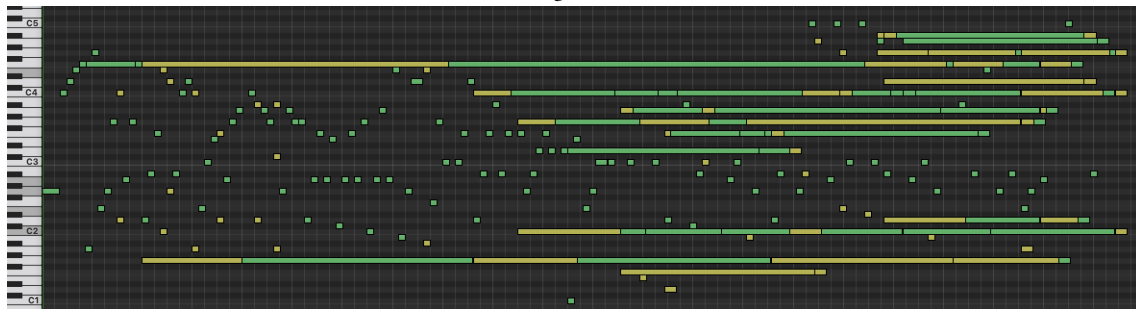
Since GPT3 is a Natural Language Processing model, music needs to be represented in a text format to be able to act as input in the model. We chose to use ABC format, because it describes music well using few characters, and there are good libraries available for converting ABC notation to MIDI and vice versa. The conversion is not lossless though, as can be seen in Figure 3.1. Figure 3.1a shows an original midi file. In Figure 3.1b the original MIDI file has been converted to ABC notation and back to MIDI using EasyABC [20] and music21 [9] APIs. In Figure 3.1c the original midi file has been converted to XML format and then ABC using music21 and xml2abc [3]. This was done without a direct function for translating MIDI to ABC, which is why the file first had to be translated into XML and then to ABC before it was recreated as MIDI.

The two recreated MIDI files both look different from the original, but the version in Figure 3.1b lies closer to the original. It contains the same notes but different note lengths, and it sounds reasonably similar when listening. The version in Figure 3.1c on the other hand, has errors in the translation of the notes which gives a dissonant result and makes it sound very different from the original.

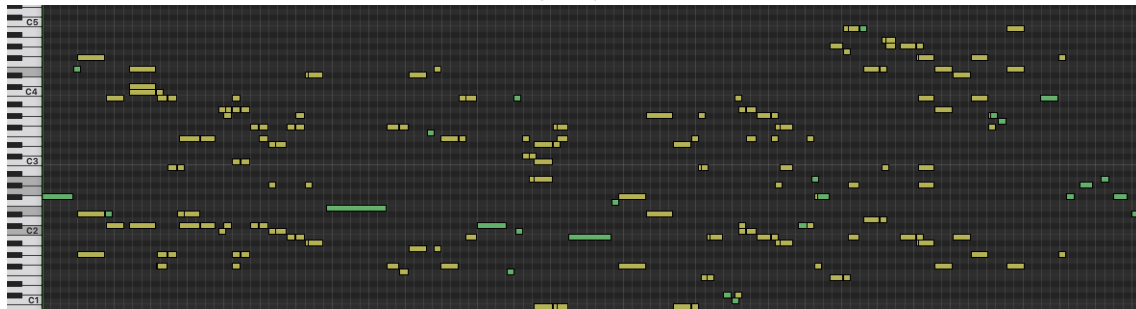
For the purpose of training, the parsing method in Figure 3.1b was chosen based on the fact that it contains the same melodic content as the original.



(a) The original midi file.



(b) Recreated using EasyABC and Music21.



(c) Recreated using xml2abc and Music21.

Figure 3.1: A song from the EMOPIA dataset (Q1_2Z9Sjl131jA_4.mid) shown both in its original version as well as after parsing to abc format and back to MIDI using EasyABC.

Chapter 4

Results

In this chapter the results of the interviews and experiments are presented. Section 4.1 presents an analysis of the interviews, Section 4.2 presents the evaluations of the data sets, Section 4.3 presents the evaluations of the pretrained models while Section 4.4 and 4.5 shows the results from retraining Performance RNN and GPT-3.

4.1 Interview results

Quotes have been translated as closely as possible from Swedish to English. The interviewees will be referred to as P1, P2, P3 and P4. The interview protocol is included in Appendix A.1.

Rhythm and structure All four interviewees were asked the question *"What would make a song really uncomfortable to listen to?"*, to try to find specific music characteristics that could be quantifiable. In discussing this, all of the interviewees highlighted the importance of rhythm and structure. A structural element that P1 and P4 mentioned is the feeling of "coming home", in a song, referring to a notion of being centered around a note from which the melodies branch, but then return to create the feeling of "coming home". When speaking of rhythm, P4 also mentioned that even though music is largely based on repetition, a negative characteristic of music could be that it is being too repetitive.

"music benefits from being organized, even unexpected things, if done with the right timing, can result in something organized and structured that we enjoy listening to", P3

"there should be a right amount of repetition, for example if something that sounds off is played over and over, some kind of repetition is created and makes it not sound off anymore. It is always a balancing act. It can also be looked at at different levels - if there is much repetition on the small scale, but none on a big scale, it is still not quite there", P4

Timbre and sound P3 and P4 mentioned that harsh and too loud sounds would be uncomfortable to listen to. P2 also touched on this subject, and said that *"there's a purely sonical aspect of music, meaning that the actual sounds we hear should appeal to us in order for us to like it"*. Timbre or sound as an evaluation parameter would primarily be applicable for models generating raw audio, but MIDI also specifies the velocity, i.e the volume, of each tone.

Background music characteristics P1 said that *"since the context is quite general, the music should be calm and act calming to the listener and not induce stress or anger"*. P2 contrasted this by saying that the music could also act as a piece of public art whose purpose is to surprise the listener. In this case the music would not necessarily have to be calming, but could be. P2 also thought that the music should suit the specific space. P4 mentioned ambient music when talking about background music, which generally means music with a slow tempo and lack of structure, commonly used for relaxation and meditation. This goes in line with the statement of P1, that background music in a public space can benefit from being calming.

4.2 Dataset evaluation results

The results from evaluating the datasets based on the chosen metrics are presented in Table 4.1. Since OpenEWLD is not in a MIDI format, we were not able to evaluate it based on compression ratio and TECs. The other data sets that were not chosen for training were not evaluated on this either.

The data sets OpenEWLD and EMOPIA are chosen for training data since their characteristics are more distinctive, which hypothetically could make it easier to deduct if a model has learnt the training data set properly. OpenEWLD and EMOPIA have a higher consonance than the other two, which means that they contain notes in the same key (see Section 2.3.1) and they have a more limited macroharmony close to the optimal range for tonal music of 5-8 notes. They also have low centricity, which indicates that they have a clearer melodic structure. The mean consonances of the data sets are also visualized in Figure 4.1.

Table 4.1: The mean and standard deviations of the chosen metrics evaluated on the chosen data sets. All of the datasets achieve a relatively high consonance.

Data set	Size	Consonance	Macroharmony	Centricity	Note Density
EMOPIA	1087	0.9572 ± 0.0603	8.4855 ± 1.5621	2.7073 ± 0.2406	0.8554 ± 0.4492
MAESTRO	1276	0.7919 ± 0.0788	11.9898 ± 0.115	3.3451 ± 0.1464	1.2942 ± 0.4459
OpenEWLD	502	0.9413 ± 0.0586	8.8805 ± 1.7428	2.7779 ± 0.2557	0.2139 ± 0.0813
TKSE	177	0.8423 ± 0.0539	11.9714 ± 0.225	3.255 ± 0.1475	0.3955 ± 0.1133
Data set	Size	Groove	Empty beat ratio	Compression ratio	TECs
EMOPIA	1087	0.9841 ± 0.079	0.0053 ± 0.0157	1.3023 ± 0.1376	27.40 ± 14.34
MAESTRO	1276	0.9761 ± 0.0084	0.0288 ± 0.0218	-	-
OpenEWLD	502	0.9308 ± 0.0271	0.0183 ± 0.0286	-	-
TKSE	177	0.9183 ± 0.0308	0.0089 ± 0.0232	-	-

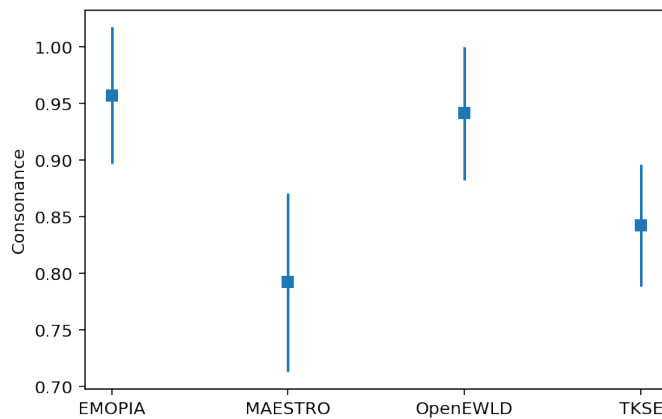


Figure 4.1: Consonance in the four datasets EMOPIA, MAESTRO, OpenEWLD and TKSE.

4.3 Pretrained model results

The four chosen generative models were run with pretrained weights and a generation temperature of 0.75. The results of evaluating 100 songs are presented in Table 4.2.

Table 4.2: The resulting evaluations of the pretrained models.

Model	Consonance	Macroharmony	Centricity	Note density
Performance RNN (performance_with_dynamics)	0.8782 ± 0.069	10.59 ± 1.613	2.8478 ± 0.3188	1.3172 ± 0.6796
Music VAE (hierarchy_dec_mel_16bar)	0.9376 ± 0.0729	8.21 ± 1.7905	2.5432 ± 0.3511	0.296 ± 0.1111
GPT-3	-	-	-	-
MuseMorphose	0.9756 ± 0.0325	8.2 ± 1.3084	2.7118 ± 0.1515	0.7935 ± 0.2596
Model	Groove	Empty beat ratio	Compression ratio	TECs
Performance RNN (performance_with_dynamics)	0.9588 ± 0.0214	0.0604 ± 0.1113	1.4457 ± 0.1699	28.8 ± 14.95
Music VAE (hierarchy_dec_mel_16bar)	0.9893 ± 0.004	0.0379 ± 0.0619	1.4781 ± 0.1541	8.06 ± 2.37
GPT-3	-	-	-	-
MuseMorphose	0.9927 ± 0.0023	0.0001 ± 0.0014	1.7478 ± 0.1521	17.38 ± 4.99

Evaluating the pretrained models gives an indication of their respective potential. The metric results do not differ that much between the models, but when it comes to modelling long term structure they perform differently.

Music VAE generated sparse samples where some were completely empty. The note density for this was much lower than for the other models, which in combination with the low note density and low number of TECs, i.e. low pattern repetition, had us discard this model. The lack of melodic structure could also be heard when listening to the samples.

MuseMorphose performed better and showed great potential, but was discarded because of a lack of documentation of how to retrain and run the model.

Performance RNN and GPT-3 were investigated further through retraining and parameter tuning. Performance RNN was chosen because it showed clear patterns and long-term structure in the generated music, and because it is easily available and is highly tuneable. GPT-3 was chosen even though it did not have any pre-trained weights for music generation. Since it is a language based model which has had great results modelling long-term structure when used in other applications, it would be of interest to try with music generation as well.

4.4 Magenta: Performance RNN

Performance RNN comes in different versions where the `performance_with_dynamics` model was chosen since it takes musical dynamics into account but is still fairly close to the base-model which is what we were interested in investigating.

The song generation command takes a seeding sequence as an input parameter, based on which the model then generates a completion. Every song generated in the same function call will therefore start with this sequence. The starting sequence is represented as a list of note values corresponding to their MIDI numbers. The MIDI number 60 corresponds to middle C, and -2 corresponds to a pause.

4.4.1 Learning rate

The model was iteratively retrained on a subset of 500 songs from EMOPIA while varying the learning rate. The learning rate was decreased from 10^{-2} to 10^{-5} by factors of 10. A batch size of 16 and the seeding sequence {C} was used for song generation. The resulting loss and accuracy for training is presented in Figure 4.2 and the generated music evaluations are presented in Table 4.3.

In Table 4.3 the music generated by the model trained using a learning rate of 0.001 received a much higher consonance score than the music generated in the other runs. It also had a lower centricity. When listening to samples from the different runs, using learning rate 0.001 gave better sounding results with better melodic structure.

The model was trained for this number of iterations due to time constraints, and it could have been trained further to clearly see which model performs the best over time. In this attempt, good scores in consonance, centricity and compression ratio and high accuracy clearly correlate as can be seen for the training with learning rate 0.001. This trained model version was therefore used in the further attempts in Section 4.4.2.

4.4.2 Seeding sequence

The impact of the seeding sequence was investigated by varying the seeding sequence for the same trained model. Training was done using a subset of 500 songs from EMOPIA, a batch size 16, learning rate 0.001 and temperature 0.75.

The results are presented in Table 4.4. There is a similar performance on all metrics apart from in note density, which is further visualized in Figure 4.3. A more sparse seeding sequence resulted in a sparser sample. This could be a direct effect of the sparse seeding sample.

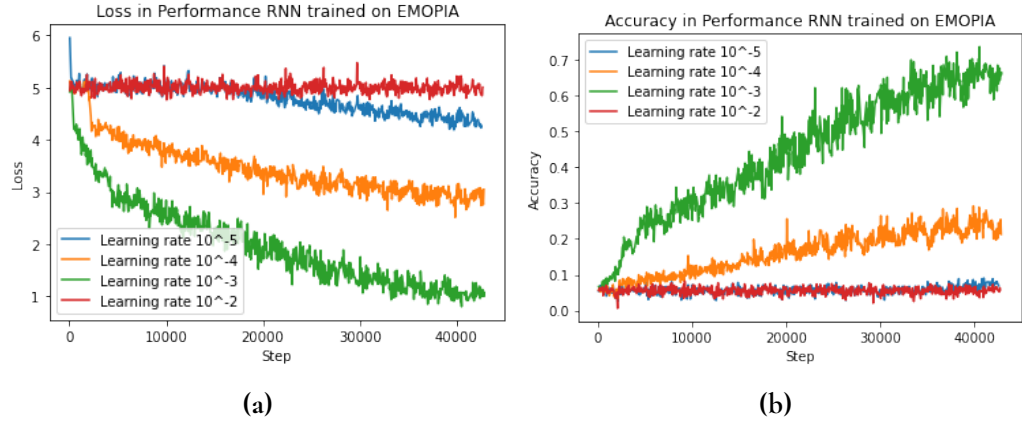


Figure 4.2: Loss and accuracy as a function of training step for re-training Performance RNN using different learning rates.

Table 4.3: The metric evaluations on music generated by Performance RNN for different learning rates.

Learning rate	Consonance	Macroharmony	Centricity	Note Density
0.00001	0.7759 ± 0.0671	11.99 ± 0.0995	3.3349 ± 0.1484	1.0429 ± 0.2909
0.0001	0.8361 ± 0.0864	11.78 ± 0.6258	3.1174 ± 0.2405	1.6938 ± 0.3352
0.001	0.9057 ± 0.0678	11.0 ± 1.1576	2.7623 ± 0.3059	1.8005 ± 0.394
0.01	0.6378 ± 0.0183	12.0 ± 0.0	3.5299 ± 0.0193	1.18 ± 0.1241
Learning rate	Groove	Empty beat ratio	Compression ratio	TECs
0.00001	0.9764 ± 0.0033	0.0005 ± 0.0028	1.3214 ± 0.0931	28.01 ± 4.68
0.0001	0.9641 ± 0.0065	0.0182 ± 0.0387	1.4561 ± 0.0452	37.35 ± 6.80
0.001	0.9646 ± 0.0067	0.0041 ± 0.0142	1.5934 ± 0.0991	33.47 ± 7.49
0.01	0.9771 ± 0.0023	0.0002 ± 0.0016	1.3591 ± 0.0280	31.55 ± 3.1571

Table 4.4: The metric evaluations on music generated by Performance RNN trained on EMOPIA for different seeding sequences. The seeding sequence is given as input at the sampling stage and acts as a starter prompt based on which the model then generates a continuation.

Seeding sequence	Consonance	Macroharmony	Centricity	Note Density
{C}	0.8537 ± 0.0814	11.66 ± 0.696	3.047 ± 0.2534	1.9185 ± 0.45
{C, E, D}	0.8828 ± 0.0756	11.37 ± 0.8906	2.9868 ± 0.2712	1.7175 ± 0.5146
{C, -, E, -, D}	0.8926 ± 0.0766	11.15 ± 1.1347	2.9239 ± 0.2786	1.6671 ± 0.5565
{C, E, -, -, D, C}	0.8871 ± 0.0729	11.28 ± 1.0685	2.9894 ± 0.258	1.6728 ± 0.4938
{C, -, E, -, -, -, D, -, C, -}	0.8799 ± 0.0709	11.22 ± 1.1277	3.0268 ± 0.2371	1.4514 ± 0.6241
Seeding sequence	Groove	Empty beat ratio	Compression ratio	TECs
{C}	0.9617 ± 0.0103	0.011 ± 0.0235	1.5847 ± 0.0906	36.9 ± 8.95
{C, E, D}	0.963 ± 0.0112	0.0161 ± 0.0412	1.5292 ± 0.0891	36.02 ± 9.79
{C, -, E, -, D}	0.9603 ± 0.0126	0.0107 ± 0.028	1.4930 ± 0.0940	37.02 ± 10.88
{C, E, -, -, D, C}	0.9594 ± 0.0118	0.0108 ± 0.0265	1.4848 ± 0.099	37.81 ± 10.24
{C, -, E, -, -, -, D, -, C, -}	0.9667 ± 0.0138	0.0123 ± 0.03	1.4472 ± 0.1163	32.59 ± 13.05

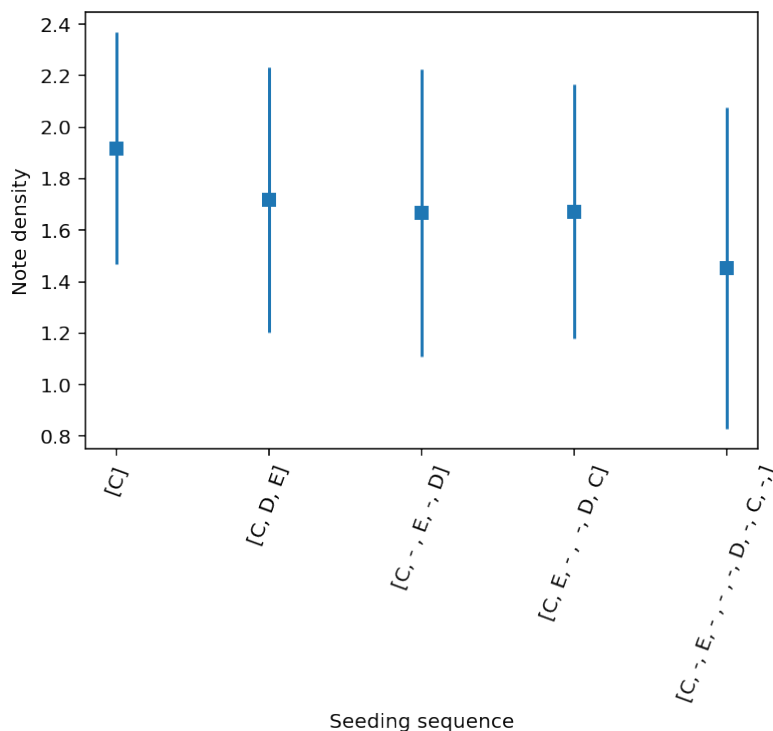


Figure 4.3: The means and standard deviations of the note density metric evaluated on songs generated with Performance RNN using different seeding sequences.

4.5 GPT-3

GPT-3 was trained on two different datasets. It was trained on a subset of 500 songs from EMOPIA, which contains songs in different keys. It was also trained on a subset of 500 songs from OpenEWLD which only contains songs in one key. OpenEWLD comes in a MusicXML format which is why it also contains information about song title and artist, which EMOPIA does not. Therefore this information can be used as song prompts in GPT-3, which is described in more detail in Section 4.5.1.

4.5.1 Prompt

GPT-3 takes a prompt as an input based on which it generates a completion. In the typical use case where it generates text, the generated response depends highly on the input prompt. The training data are arranged in these prompt-completion pairs from which the model learns. When instead training to generate music, what the input prompt should contain is not as evident. We will attempt two setups to investigate the impact the prompt has on the result.

Compared to the text generation use case, the same prompt should be able to give different completions, e.g. "How are you?" could be completed with multiple different answers, "Great, how are you?", "Not to well actually.", "Good." etc. Therefore the song prompts do not necessarily have to be unique, since many different completions can be plausible for one prompt.

OpenEWLD

In the first setup the model was trained on the OpenEWLD dataset which contains information about artist and song titles for each song. The artist and song title were used as unique prompts for each song input to the model. Each song in this dataset was transposed to the same key before used as input in training.

OpenEWLD example prompt:

"X: 1 \$ T: A song about trees \$ C: The oaks \$ <song>"

EMOPIA

In the second setup the model was trained on the EMOPIA dataset which does not contain any information about artist or title. Instead the ABC tune header was used as input. The ABC header contains song-specific information regarding time and key, which has an impact on the output of the song. These prompts are therefore not unique for each song. This was investigated both for the dataset transposed to the same key as well as kept in their original keys.

EMOPIA example prompt:

"X: 1 \$ M: 3/4 \$ L: 1/16 \$ K: Em \$ <song>"

4.5.2 Learning rate

The impact of the learning rate multiplier (lrn) in GPT-3 was investigated by training the model for 7 epochs for the lrn 0.02, 0.08, 0.14 and 0.20. The default learning rate is multiplied by the lrn, meaning that a smaller lrn gives a smaller resulting learning rate. The loss and accuracy are presented in Figure 4.4.

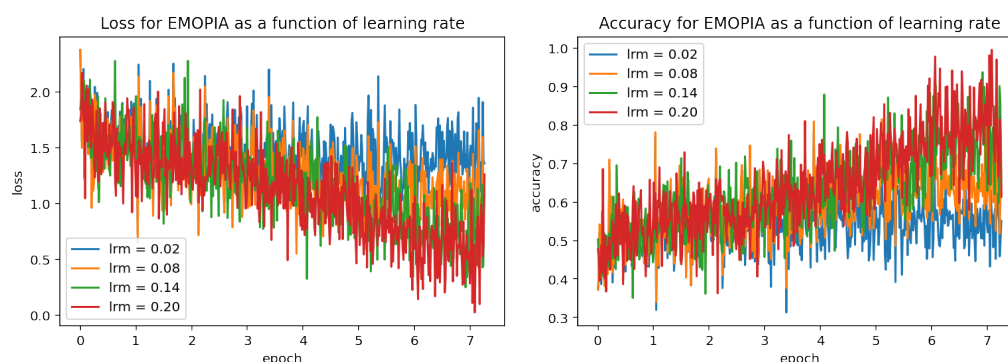


Figure 4.4: The loss and accuracy for GPT-3 trained on a subset of EMOPIA for 7 epochs.

Training should be carried out for more epochs in order to be able to draw a conclusion of optimal learning rate. As expected, a lower learning rate implies slower learning. The highest learning rate with an lrn of 0.2 gave a relatively high accuracy already after 7 epochs. This would be interesting to investigate further.

4.5.3 Epochs

The impact of the training length was investigated by using a high number of training epochs. A batch size of 32 and a learning rate multiplier of 0.2 was used. The resulting loss and accuracy are presented in Figure 4.5 and the evaluation of the generated songs are presented in comparison with the training data in Table 4.5. The evaluation scores of generated music ends up close to the scores of the training data, and achieves both a high consonance, limited macroharmony and low centricity. It does not contain as many TECs, i.e repeated patterns, as the training data, but since the standard deviation of the number of TECs in the training data is high this does not have to mean that the generated music lacks structure. Listening to the music also confirms that it contains melodic structure.

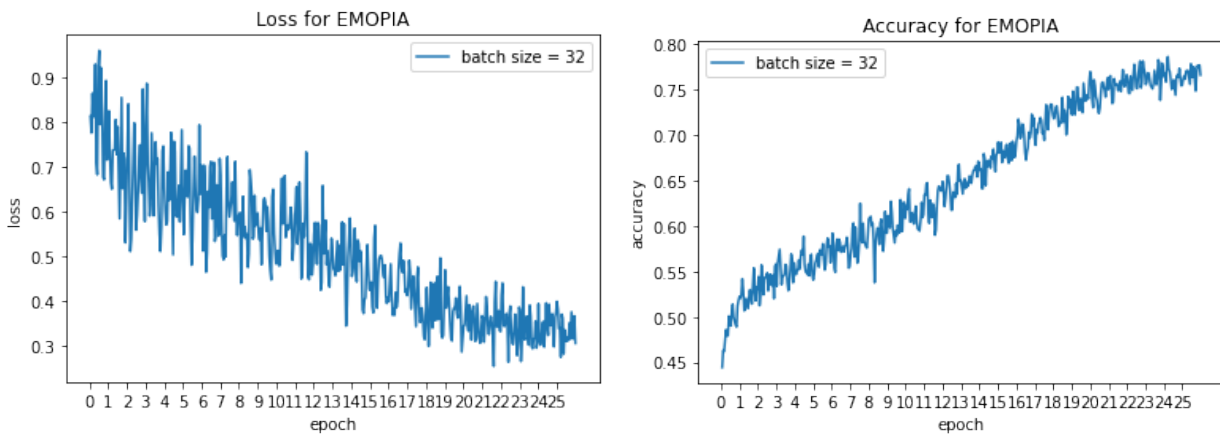


Figure 4.5: The loss and accuracy for GPT-3 trained on a subset of EMOPIA for 25 epochs.

Table 4.5: Evaluation of EMOPIA and GPT-3 trained on EMOPIA for 25 epochs.

Data set	Consonance	Macroharmony	Centricity	Note Density
EMOPIA	0.9572 ± 0.0603	8.4855 ± 1.5621	2.7073 ± 0.2406	0.8554 ± 0.4492
GPT-3, trained on EMOPIA, 25 epochs	0.998 ± 0.0074	6.6566 ± 0.9657	2.3309 ± 0.3972	1.3088 ± 0.8328
Data set	Groove	Empty beat ratio	Compression ratio	TECs
EMOPIA	0.9841 ± 0.079	0.0053 ± 0.0157	1.3023 ± 0.1376	27.40 ± 14.34
GPT-3, trained on EMOPIA, 25 epochs	0.9966 ± 0.0014	0.0 ± 0.0	1.9304 ± 0.2381	16.37 ± 4.44

The loss and accuracy for a training iteration using OpenEWLD is shown in Figure 4.6 for comparison. The same training parameters were used and the only difference is the training data. Here where the model reaches a higher accuracy than in Figure 4.5 already after 3 epochs. This implies that the data set impacts the training results.

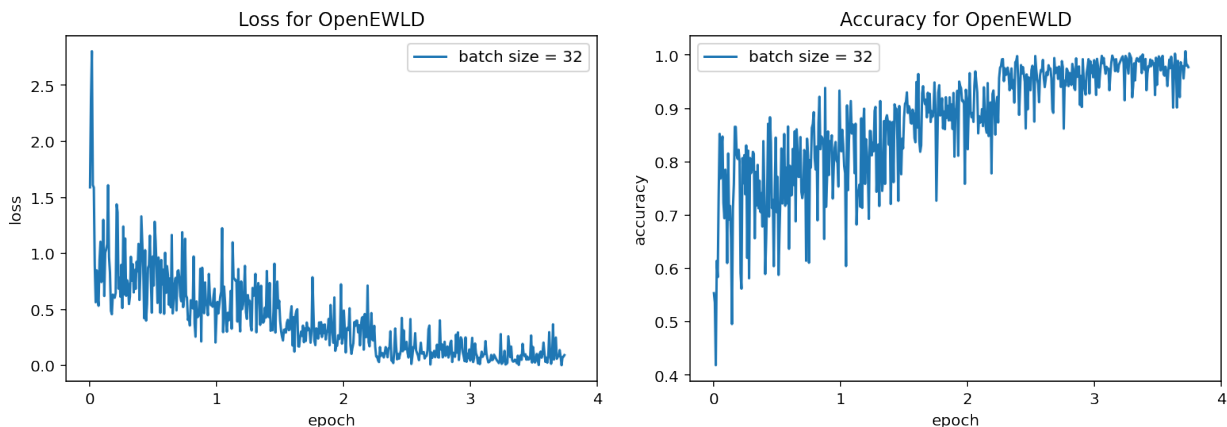


Figure 4.6: The loss and accuracy for GPT-3 trained on a subset of OpenEWLD for 4 epochs.

4.5.4 Transposed training data

To investigate the impact of homogeneity in the training data GPT-3 was trained on both the unchanged EMOPIA dataset as well as the same data set where every song was transposed into the same key. Figure 4.7 shows the key content of the unaltered EMOPIA dataset. The EMOPIA dataset was divided into major and minor scales and thereafter transposed to either C minor or C major. The model was then trained on the minor and major data sets separately. A temperature of 0.75 was used for sample generation. The biggest difference in evaluation results in Table 4.6 can be seen in note density, where it is much lower for the training iterations using the homogenous datasets. A difference can also be seen between the smaller data set, EMOPIA Major, and the two others, which gets a higher macroharmony and a higher number of TECs.

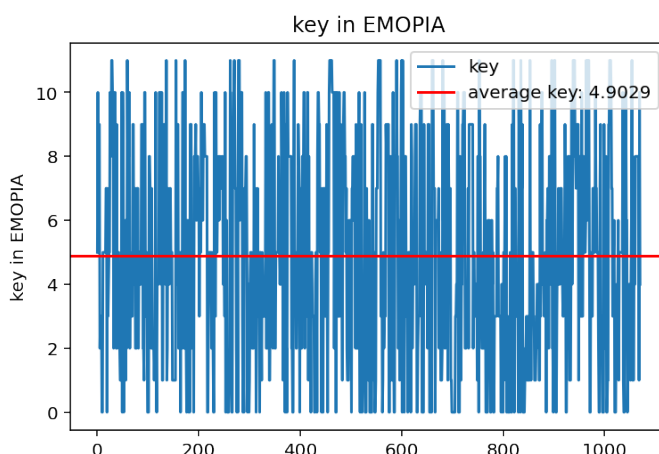


Figure 4.7: How the songs in the EMOPIA dataset are distributed over keys.

Table 4.6: Metric evaluations for GPT-3 trained on regular EMOPIA and transposed EMOPIA.

Data set	Size	Consonance	Macroharmony	Centricity	Note density
Standard EMOPIA	500	0.9962 ± 0.0191	5.9588 ± 1.399	2.0365 ± 0.5558	1.6951 ± 4.2858
EMOPIA Major	405	0.9943 ± 0.0153	6.45 ± 1.5058	2.0663 ± 0.4744	0.8717 ± 0.508
EMOPIA Minor	500	0.9981 ± 0.0078	5.8774 ± 1.3716	1.9658 ± 0.6256	0.9882 ± 0.5715
Data set	Size	Groove	Empty beat ratio	Compression ratio	TECs
Standard EMOPIA	500	0.9963 ± 0.0012	0.0 ± 0.0	2.2459 ± 0.5366	13.52 ± 4.86
EMOPIA Major	405	0.9961 ± 0.0012	0.0 ± 0.0	2.103 ± 0.4628	17.84 ± 5.78
EMOPIA Minor	500	0.9958 ± 0.0014	0.0 ± 0.0	2.2509 ± 0.54	13.41 ± 4.81

Table 4.7: Metric evaluations for GPT-3 for varying generation temperatures.

Data set	Temperature	Consonance	Macroharmony	Centricity	Note density
OpenEWLD	0.5	0.9141 ± 0.0273	10.09 ± 0.8258	2.9736 ± 0.0644	0.379 ± 0.0473
OpenEWLD	0.75	0.9478 ± 0.0369	9.54 ± 1.4793	2.8809 ± 0.1604	0.4412 ± 0.1557
OpenEWLD	1	0.9362 ± 0.0489	10.03 ± 1.5777	2.9387 ± 0.1804	0.494 ± 0.1639
EMOPIA	0.5	0.9979 ± 0.0124	5.6289 ± 1.3646	1.8907 ± 0.5681	1.5119 ± 1.9672
EMOPIA	0.75	0.9954 ± 0.012	6.8021 ± 0.9199	2.3056 ± 0.4132	1.2118 ± 0.6679
EMOPIA	1	0.988 ± 0.0238	7.5747 ± 1.2561	2.5225 ± 0.3829	1.0842 ± 0.6821
Data set	Temperature	Groove	Empty beat ratio	Compression ratio	TECs
OpenEWLD	0.5	0.9992 ± 0.0001	0.0 ± 0.0	2.9972 ± 0.5308	7.02 ± 1.56
OpenEWLD	0.75	0.9985 ± 0.0006	0.0055 ± 0.0133	3.2550 ± 1.3071	7.85 ± 2.77
OpenEWLD	1	0.9983 ± 0.0007	0.0047 ± 0.0136	3.1388 ± 1.0121	9.06 ± 4.00
EMOPIA	0.5	0.9964 ± 0.0011	0.0 ± 0.0	2.3899 ± 0.6735	13.19 ± 4.62
EMOPIA	0.75	0.9967 ± 0.0011	0.0 ± 0.0	1.9278 ± 0.2485	15.80 ± 4.32
EMOPIA	1	0.9969 ± 0.0011	0.0003 ± 0.0023	1.7051 ± 0.1655	18.08 ± 4.61

4.5.5 Temperature

In Table 4.7 the results from varying the temperature are presented. The same temperature variations were made for both the model trained on the OpenEWLD and the EMOPIA data sets. The same prompt was used for generating music from both models:

Prompt :

"X: 1 \$ M: 4/4 \$ L: 1/4 \$ K: C \$ <song>"

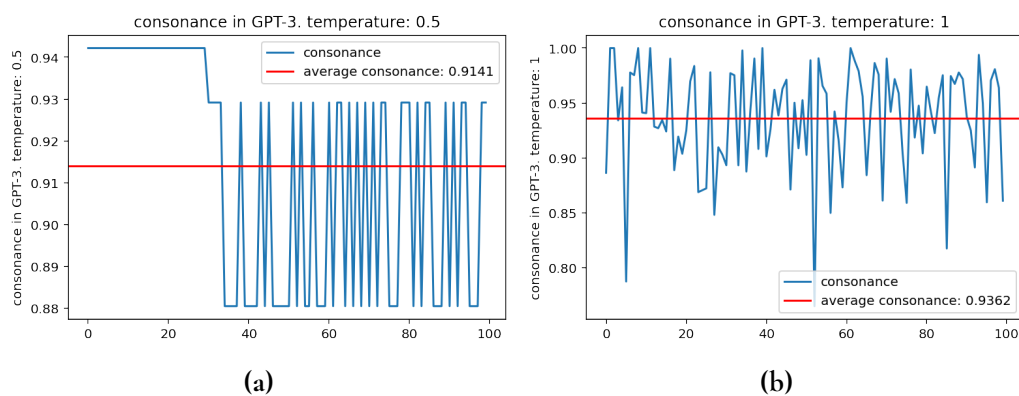


Figure 4.8: The consonance in two different sample sets generated using GPT-3 retrained on OpenEWLD. A temperature of 0.5 and 1 were used respectively. The consonance of the two different generated data sets showcases that a low temperature gives low diversity. The same sample is generated multiple times and the whole data set contains only 3 unique samples.

Chapter 5

Discussion

In our experiments, two sequential generative models, GPT-3 and Performance RNN were evaluated on two different data sets, OpenEWLD and EMOPIA. In this section the experiment setup and the results of the experiments are analysed. Conclusions are also made in relation to the research questions.

5.1 Model performances

This section discusses different aspects of the training results and the differences between the two models.

5.1.1 Data set homogeneity

We found that the key of the data set had an impact on the note density as can be seen in Section 4.5.4 and Table 4.6. Transposing the dataset to the same key gave a note density closer to the one of the training data set. This indicates that conforming the data set to only contain the same key makes it easier for the model to learn the patterns in it. The note density was still much higher in the generated music than in the training data, which was seen consistently throughout all experiments. It is not certain what causes this or if it is an indicator of the music sounding bad. It can also be said that when the author listened to the music with varying note densities, there is no particular note density that sounds better than the other quality wise, but there is a difference in how stress-inducing the music sounds. This would be interesting to investigate further with listener-studies to be able to draw conclusions about a potential correlation between the calming effect and note density.

There could also be differences seen between the training with the bigger and the smaller data sets. Since the EMOPIA dataset naturally contains fewer songs in major keys than in minor keys training was done on different data set sizes. The model that was trained on a smaller data set had a larger macroharmony score as well as more TECs, i.e more repeated

patterns. This could both be from the model being slightly less trained than the other two, but could also be a result of melodic differences that were present in these particular subsets of EMOPIA. Repeating this attempt using more data could clearer show what impact this has on the results, but we can already with these small data set sizes see an indication in the pattern metrics that the model trained on the smaller data set had not learnt the structure patterns as well as the other models.

5.1.2 Learning rate

Both the attempts at varying the learning rate for the two models in Section 4.3 and Section 4.5.2 would have to be run for more epochs in order to find the most optimal learning rate value. In this case, the optimal learning rate for a particular running time was found, which was helpful in achieving the best results for the running times available in this project, but the found learning rates does not have to be the best for longer running times.

5.1.3 Epochs

Training GPT-3 with EMOPIA and OpenEWLD achieved very different accuracies as seen in Figure 4.6 and Figure 4.5, and training on OpenEWLD achieved a higher accuracy than EMOPIA with a fifth of the training time. As discussed in the previous section, the optimal learning rate for longer training times does not have to correspond to the learning rate used, which could have limited the quality of the results. Since the same learning rate was used in training with both datasets, this indicates that training is highly dependent on the contents of the data set and that parameter tuning is required for every data set individually.

5.1.4 Temperature

Investigating the impact the temperature had on GPT-3 showed that the evaluation metrics are not in themselves a clear indicator of a good generated data set when we are specifically interested in a diverse data set. In Figure 4.8, both generated data sets have a high consonance, i.e how much of the song is in the same key, and contain songs that sound good to the author when listening, but if the model is to be used to generate varied samples to play in a public space the results in Figure 4.8b are preferred over the results in Figure 4.8a. For low temperature the model generated almost the same song over and over. This follows the expected behaviour of using a low temperature which inherently lowers the diversity and makes it only generate the sample with the highest probabilities, i.e. the same over and over if the prompt is the same. In the example shown in Figure 4.8 the songs were generated using the same prompt, which is why an alternative idea to achieve more diverse samples could be to vary the prompts in a smart way by for example randomizing song titles as prompts.

5.1.5 Data format parsing

When retraining GPT3, the MIDI files in the training data set were parsed from MIDI to ABC notation. This conversion between different file formats always comes with some loss of information due to the nature of the formats. A conversion also happens in Performance

RNN, where the MIDI files are converted into a vector representation to be able to be used as input in the LSTM network. This has a big impact the outcome of the training since the original data set can be assumed to have been altered in some way from parsing. This aspect is difficult to eliminate completely, but the impact of it could be made as small as possible by choosing file formats that do not require many parsing steps. Furthermore, the two different datasets that GPT-3 were retrained on came in different formats, MusicXML and MIDI, which results in a difference in parsing between the two runs with the same model. It would therefore be interesting to find a way of measuring the information loss and the extent of the alteration of the training data in order to include this in the comparison between the two models as well as the runs with different datasets using the same model.

5.2 Metric

Discussing music quality and fundamental demands on music in the interviews gave a good understanding about what parameters that would be interesting to quantify in an evaluation metric. It also showed some of the diversity in liking and opinions, and the difficulty in summarizing the broad spectrum of musicality into numbers. Because of the creative and artistic perspectives on music, it can be a difficult task to classify something as "good" or "bad" since there is always someone that is going to disagree. The following sections evaluates the different metrics and how well they encapsulate music quality.

5.2.1 Note density

P1 stated in their interview that the music should have a calming effect on the listener. As mentioned in Section 3.3, previous studies [5] [16] have shown that a low song tempo lowers the heart rate which makes us relaxed. Therefore the note density of a song could potentially have a correlation with the calming effect of the music, since the note density is also changed with tempo. From listening to the music, the note density did not seem to have a correlation with the quality or musicality of the samples.

5.2.2 Empty beat ratio

This metric showed little variation over the resulting evaluations. Instead of calculating the mean of a set of songs, this metric could be used to sort out songs that contain too much silence.

5.2.3 Consonance

The consonance metric is based on what ratio of the notes in the song belong to its most common key. The most common key is found by parsing through all of the notes and finding which belong to what key. Since parallel keys consist of the same tones, they get the same scores. Therefore there is a bias towards the key that is parsed first, which can lead to the key being misdetermined. This happens when GPT-3 is trained on the OpenEWLD dataset. All generated songs ideally use the tones in the key C, and therefore some are found to be

in its parallel key, Am, which lowers the consonance metric. By improving the key-finding algorithm and taking chord progressions and melodic structure into account, this could be improved.

5.2.4 Centricity

The centricity metric, which measures if some notes are more common than others, could be an indicator of the music quality. A lower value of this metric means that some notes are more common, which indicates that the model does not choose notes undeliberately but has instead learnt the harmonic patterns that are common in music. The scores varied between 2 and 3, which corresponds to the scores of the data sets, i.e. of real music. The training iterations that achieved higher accuracy also achieved a lower centricity score. For example this can be seen in Table 4.3, where the model trained with the learning rate that achieved the highest accuracy also obtained the lowest centricity score. When listening to samples with different centricity scores, it is difficult to distinguish a difference due to the difference in music content in the training data sets as well as other quality parameters that varied. This would therefore be interesting to further correlate with human perception.

5.2.5 TECs

It showed to be difficult to quantify structure and repeatable patterns such as motifs or chord progressions in one number. Our attempt was to calculate the number of TECs, but what a "good" number of TECs is is not completely clear as this can vary immensely between songs depending on music genre and song length for example. It could be said though, that a very low number of TECs which means that there are no or very few repeated patterns in the song, would be undesirable. For example, when listening to the music generated by the pre-trained VAE model MusicVAE (see Section 4.3), the music clearly sounded disorganized and structure-less, which corresponds to the low number of TECs found in Table 4.2. An indication of the performance could therefore be given by comparing the number of TECs to the mean number of TECs of the training data. As seen in Table 4.5 the EMOPIA dataset varied a lot in the number of TECs with a standard deviation of ~50% of the mean value which complicates this by not having a strict interval.

There is also the additional aspect that potential sources of error here could stem from the SIA algorithm's capability of finding the repeated patterns in the song. The COSIATEC implementation of SIA (see Section 2.3.1) that was used only allows a tone to be part of one repeated pattern [15], which is why there could be an underdetermination of patterns.

5.2.6 Macroharmony

The macroharmony metric shows how well the model knows how to arrange the notes. If the macroharmony uses all the possible notes, i.e. 12, it could be presumed that the music is not staying within one key. In Table 4.4 where the seeding sequence was varied in Performance RNN, the macroharmony is close to 12, and the resulting music also sounds incoherent and messy. On the contrary, in Table 4.6 and in Table 4.7 where temperature and data set key content were varied the macroharmonies were lower. This indicates that the music content

of the generated songs is not just randomized, and instead follows a clear pattern that the model learned. This metric is a clear indicator of the model performing well.

5.2.7 Groove

The groove metric, measuring the consistency of rhythm throughout the music, scored similarly for all the music that was evaluated. Therefore not many conclusions can be drawn from this metric, and instead this could maybe be more useful when evaluating music with rhythmic instruments in them, as rhythmic patterns could be more easily distinguishable and possibly more erroneous.

5.3 Legal aspects

In the field of AI-generated music, the legal question of copyright infringement arises. There is a difference between being inspired by someone else's work and directly copying it, but where AI falls in this matter is a current topic of discussion where the laws and rules are not completely clear. Sampling of music, i.e. using part of someone else's song in your own, can be an infringement of copyright and therefore illegal [2]. This specific phenomenon of sampling music has laid ground for well-known court cases in the past [22]. Copyright means that the song creator owns the exclusive right to distribute, perform and make new copies of the work in question [2]. Training an AI-model on copyrighted data is generally a grey-area, where copyright infringement both could happen when using a copyrighted song in training, or if the generated music ends up sounding too similar to the original training data.

In the first case, the process of training an AI model is not all that different from the human learning process. Our lives are a constant processing of data, meaning that we learn about and categorize impressions and influences we experience in our daily lives. Therefore everything we create is implicitly inspired by our surroundings to some extent, even if we maybe are not aware of it. So is it labeled copyright infringement, if someone listens to the Rolling Stones their whole life and then writes a rock song? The line between inspiration and plagiarism can be blurry for human composers, and the same goes for the world of computer-made music.

In the second case, deliberately trying to create a song in the style of a specific artist could be problematic, though still a grey-area. If it would sound identical to a specific song of the artist, it would most certainly be a possible copyright infringement. This scenario could be difficult to rule out, and if the AI generates hundreds or thousands of songs, some could resemble the training data identically. In an autoencoder model a sample could be taken exactly from a training data point in the latent space, resulting in a song that already exists.

In practice it would be complicated to prove which songs a neural network has been trained on, since the evidence of it is only the trained weights of the network. Reverse engineering millions of weights to find the initial training data would be practically impossible.

Chapter 6

Conclusion

This thesis researched generative AI models for music generation. By revisiting the research questions multiple conclusions can be drawn.

RQ1 Which already developed models in the state-of-the-art can be utilised to achieve music generation?

By evaluating the training process based on training accuracy and loss, as well as evaluating the generated music using the custom metric, we found that transformer and LSTM models work well for generating music. We also conclude that they perform well in generating samples with long-term structure.

RQ2 What are the demands on the quality of background music in public spaces?

Through the interviews a clear set of important characteristics were determined which showed to be useful later in the work. The interviews also showed the diversity in music and the difficulty of quantifying such a varying form of media.

RQ3 How can the quality of generated music be measured?

We found a set of metrics, where some clearly correlated with music quality, and where some did not vary much between music of varying quality. Thus, we conclude that there are many aspects of accurately modeling the quality of music, and that a quantitative metric should be correlated with human perception in order to obtain meaningful limits.

RQ4 To what extent can the existing generative models of background music generate background music suitable for public spaces?

Both GPT-3 and Performance RNN generated music of acceptable quality. They are both affordable to train in terms of time, and generate music with clear structural elements and melodic contents that follow the conventions of music to a large extent. They both implement model architectures, transformer and LSTM, that have inherent

sense of context which this work also confirmed.

We found that the data set had an impact on the training time as well as on the parameter setup needed to achieve good results. Additionally, the parameter setup at generation also impacted the outcome of the music. Finally we conclude that the generation of licence free music using AI is a feasible method which gives usable results.

Future work

In future work it would be of interest to correlate human perception of music with the numerical metrics used in this thesis. This would give clear indications of what scores to aim for in the metrics.

- It would also be interesting to further investigate ways of measuring repetitions and patterns in music and finding a metric that summarizes the structure in a song in a meaningful way.
- Since this thesis project was limited by time and computing resources, another improvement would be to train the models for longer and evaluate the differences.
- Furthermore, it would be beneficial to find in what ways the data set contents affect the outcome of training, i.e what music is easier to train on.

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Appendix A

A.1 Interview documents

A.1.1 Consent to be interviewed for research

Consent to be Interviewed for Thesis Project

Project: Generating Music using AI

M. Sc. Thesis Student: Ebba Rickard (ebba.rickard@gmail.com)

Supervisors: Emma Söderberg (emma.soderberg@cs.lth.se), Johan Davidsson (johan.davidsson@axis.com), Danny Smith (danny.smith@axis.com)

Purpose: The purpose of the thesis project is to investigate different machine learning methods that can be used to generate music.

Participant's role in the study: Participate in interview about music quality

Responsibility of the interviewer

Before the interview the interviewer will have:

- Explained the purpose of the interview and how any interview answers or personal information will be handled.
- Explained the rights of the participant, including the right to withdraw at any time.

What will happen to the interview results

The interview will be recorded and kept confidential and only made available to the interviewer and the supervisors of the project. Parts of the interview answers may be used in the project report, but under no circumstances will personally identifiable information be included, unless the participant explicitly wants to not be anonymous.

Rights of the interviewee

I understand the purpose of this interview as explained to me by the interviewer and I agree that I can withdraw from the interview at any time. If I choose to do so, I understand that any potential data already collected will be removed. I understand that I can choose to decline to answer any questions.

I agree that my answers during the interview will be recorded and that this data will be handled in accordance with the section above.

.....
Date

.....
Interviewer Name

.....
Interviewer Signature

.....
Interviewee Name

.....
Interviewee Signature

A.1.2 Interview Protocol

Context

The aim of this master thesis project is to generate music using machine learning models. Since computers don't know anything about anything until we teach them, it's important to break down the concept of music into its most basic parts in order for the computer to be able to understand it. Therefore, facts that us humans might think of as obvious also need to be defined carefully. In addition to this, knowledge about music theory could be used to automate the evaluation process of the results in a simple way.

Consent

It is up to you if you want to answer a question or not, and the interviewer is always available to answer questions after the interview. The interview will be recorded in order to simplify the interview process and the evaluation of the answers. The recording will be shared with the supervisors of the project, and will be discarded after the project is finished. The answers in the interview could be included in the report, but in this case they will be completely anonymous. The final report will be shared with the interview subject.

Interview questions

General

What is your background in composition? Music theory?

Vad har du för bakgrund inom komposition? Musikteori?

If applicable, what was your process the last time you composed a song? Do you have a common strategy that you use? Do you decide on tempo etc beforehand or does it happen organically?

Om du har någon, vad var din strategi den senaste gången du komponerade? Har du en generell strategi när du komponerar musik? Bestämmer du taktart och tempo osv i förväg eller sker det organiskt?

(In what way does the music theory you know affect the music you create?

På vilket sätt påverkar din teoretiska kunskap den musik du skapar?)

What factors do you think could contribute to making a song really uncomfortable to listen to? Structure? Sound? (For you but also for the average person, the big mass)

Vilka faktorer tror du skulle kunna bidra till att göra musik riktigt obekväm att lyssna på?

(Både för dig men också för gemene man, den stora massan)

What kind of music do you like to listen to? What do you think it is that makes this music enjoyable?

Vilken musik brukar du lyssna på? Vad tror du det är som gör den musiken "bra"/bekväm att lyssna på?

Limit one parameter e.g. melody, can only use one single tone, how would you compose it to make it musical?

Begränsa en parameter ex melodi, får bara använda en enda ton, hur skulle du komponera för att göra det musikaliskt?

Kortfattat:

According to you, what theoretical demands are there on music for it to be comfortable to listen to? (Rhythm, speed, key, melody, chords, singing, what instruments, genre, timbre)
Enligt dig, vilka musikteoretiska krav finns på musik generellt för att den ska vara bekväm att lyssna på? Taktart? Tempo? Tonart? Melodik, ackordföljder? Sång, vilka instrument? Genre? Klangfärg?

Project context, background music

What type of music would you want to put in this kind of context? (A very general context where the listener is unspecified and could be practically anyone)

I kontexten, vad tror du skulle vara bra musik att spela? (En väldigt generell publik miljö med en ospecificerad lyssnare)

How would you define background music?

Hur skulle du definiera bakgrundsmusik?

What would you consider bad background music?

Vad skulle du tycka är dålig bakgrundsmusik?

If applicable: How is your thought process when composing film music compared to composing a song with the purpose to be listened to independently? How would you approach composing background music?

Om relevant: Hur tänker du när du komponerar musik till film jämfört med när du komponerar en låt som är tänkt att lyssnas på fristående? Hur hade du tänkt om du skulle komponera bakgrundsmusik?

Generated music

This part of the interview will consist of listening to a couple of AI generated songs and trying to find the factors that make them sound "fake" or robot-like.

EXAMENSARBETE Generating music using AI**STUDENT** Ebba Rickard**HANDLEDARE** Emma Söderberg (LTH), Johan Davidsson (Axis), Danny Smith (Axis)**EXAMINATOR** Elin Anna Topp (LTH)

Generera musik med AI

POPULÄRVETENSKAPLIG SAMMANFATTNING **Ebba Rickard**

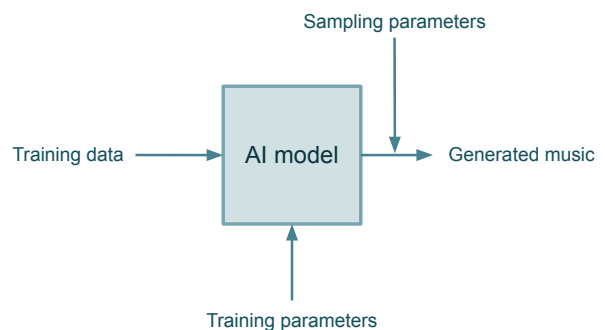
AI i rollen som upphovsmakare av kultur är ett relativt nytt fenomen. Detta examensarbete utforskar specifikt skapandet av musik med hjälp av AI-metoder.

För att spela musik i offentliga utrymmen krävs det att man följer gällande regler kring upphovsrätt och då innehar en kommersiell licens. Som ett alternativ för butiksägare och andra aktörer som spelar musik i offentliga utrymmen skulle en möjlighet kunna vara att erbjuda licensfri musik. Musiken skulle kunna genereras av AI, då AI-genererad musik är licensfri om modellen tränas på helt upphovsrättsfri musik.

Musiken skulle kunna vara licensfri även om detta frångås, men juridiken kring kultur genererad av AI är omdiskuterad på grund av att det är ett nytt och till stor del oprövat område. Diskussionen handlar dels om frågan om vem som är upphovsmakare till verken som skapas, men också om frågan om vilka verk som får användas i träningen av AI-modeller. Jämförelsen görs ofta med människans kreativa process, där musiker lyssnar på musik med upphovsrätt, och sedan skapar egen musik, oundvikligen influerade av de verk den lyssnat på i livet.

I detta examensarbete undersökte vi möjligheterna att generera alternativ, licensfri bakgrundsmusik med hjälp av maskininlärningsmetoder. Två existerande generativa AI-modeller utvärderades och tränades om. Vi valde att undersöka Long Short Term Memory-modellen (LSTM) Performance RNN från Magenta Tensorflow samt transformermodellen GPT-3 från OpenAI. Båda valdes baserat på att deras bakomliggande strukturer är speciellt bra på att modellera sekventiell

data. Den förstnämnda är gjord för att generera musik i MIDI-format medan den andra är en stor språkmodell som kan användas till att generera musik som representeras i ett språkbaserat format. Musiken utvärderades med hjälp av en handfull olika kvantitativa mått baserat på teori om musik, teori om tonalitet samt om musiks matematiska struktur.



Resultaten av försöken visar att träningsdatans strukturella innehåll hade stor inverkan på hur många iterationer som krävdes för att modellerna skulle uppnå hög träffsäkerhet. Specifikt spelade det roll om all musik i träningsdatan gick i samma tonart eller inte, samt hur hög deras notdensitet var. Dessutom spelade parametriseringen i samplingssteget roll för att kunna utnyttja den färdigtränade modellens fulla potential. Båda modellerna lyckades modellera övergripande struktur i musiken.